

- 11** **C** For geostationary satellites, all satellites orbit the Earth at a fixed distance from the Earth, r , and the orbital period T is the same as Earth's rotation of 24 hours. Hence, all geostationary satellites also have the same angular velocity by

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$$\omega = \frac{2\pi}{T} \text{ (options A and D are eliminated)}$$

For a satellite, the gravitational force F_G provides the centripetal force. Therefore,

$$\frac{GMm}{r^2} = mr\omega^2$$
$$\frac{GM}{r^2} = r\omega^2 = g$$

Therefore, the centripetal acceleration which is dependent on the radius and orbital period is the same for geostationary satellites. **(Option B is eliminated)**

Even though the speed of the satellites are the same, as given by the equation $v = r\omega$, the mass of the satellites may be different. So by $KE = \frac{1}{2}mv^2$, the kinetic energy of the satellites may be different. **(Option C is the answer)**